SALIENCY-GUIDED VISION TRANSFORMER FOR FEW-SHOT KEYPOINT DETECTION

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Paper under double-blind review

ABSTRACT

One attractive property of Vision Transformer (ViT) is to capture long-range dependency among image patches, which helps improve few-shot keypoint detection (FSKD) and is not explored yet in the literature. However, the simple application of ViT brings in irrelevant features outside of the region of interest due to the global attention matrix, thus degrading similarity learning between support and query features. In this paper, we present a novel saliency-guided vision transformer, dubbed *SalViT*, for few-shot keypoint detection. Our SalViT enjoys a uniquely designed masked self-attention and a morphology learner, where the former introduces saliency map as a soft mask into ViT and constrains the self-attention to foregrounds, while the latter leverages the so-called power normalization adjusting morphology of saliency map, realizing dynamically changing receptive field. With the SalViT, we explore the use of ViT features together with CNN features to model both local and long-range dependency, providing more informative representations for similarity learning. We apply SalViT to FSKD in inductive and transductive settings, and show that it outperforms other methods.

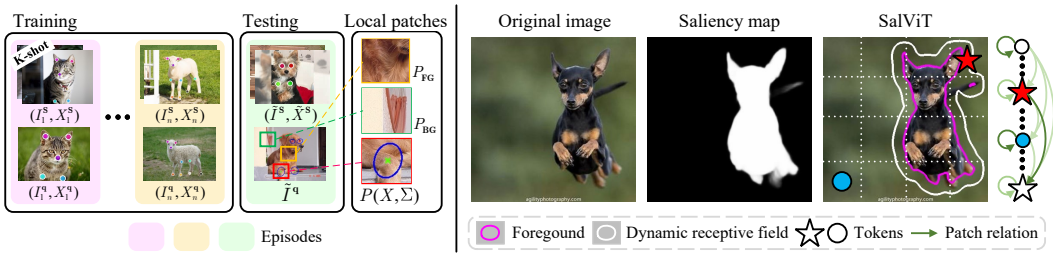
1 INTRODUCTION

The deep fully-supervised learning has several weaknesses, *e.g*. heavily relying on large amounts of labelled data for training and poor generalization. In contrast, humans can learn new concepts quickly given a few demonstrations, which motivates the need for few-shot learning (FSL) approaches (Wang et al., 2020b; Sung et al., 2018; Koch et al., 2015) in various tasks such as image classification, object detection, segmentation, and keypoint detection. The emerging few-shot keypoint detection (FSKD) (Lu & Koniusz, 2022) localizes keypoints in query images given one/few support images with keypoint annotations (Figure 1(a)). The hardest setting (which we use among other settings) due to the domain shift in keypoints and species of FSKD is that both the base keypoints and species in training are *disjoint* with the novel ones in testing.

FSKD faces two common challenges: 1) generalization to unseen classes; 2) low number of samples. Though seminal works (Vinyals et al., 2016; Snell et al., 2017; Sung et al., 2018) address some of such issues, FSKD faces additional challenges. Firstly, keypoints are some semantic local regions that are smaller/harder to detect than holistic objects. Secondly, as shown in Figure 1(a), the similar local patterns (*i.e*., noise) from foreground and background affect FSKD. Moreover, predicted and groundtruth keypoints inherently exhibit ambiguity of location. While modeling both localization and semantic uncertainty (Lu & Koniusz, 2022) helps, the interference under similar local patterns and noise challenges FSKD. We note that each keypoint has close semantic and spatial relationships with other foreground patches, *e.g*. left-eye lying left to right-eye and above on shoulder. Such relations require context modeling. Thus, we argue that the long-range foreground relations are important in FSKD as they help infer more accurate keypoints.

Vision transformer (ViT) (Dosovitskiy et al., 2021) is a natural choice to capture such long-range relations. However, the global receptive field of self-attention (SA) in ViT leads to costly computations, and introduces irrelevant features outside of foregrounds (Xia et al., 2022). Thus, we propose to use saliency maps to let SA focus on foregrounds (Figure 1(b)) and ignore background noise.

In this paper, we present a saliency-guided vision transformer (SalViT) for few-shot keypoint detection (FSKD). The SalViT captures the long-range dependency within salient regions to build infor-



(a) Few-shot Keypoint Detection (b) Patch Relation Learning with SalViT

Figure 1: Illustration of few-shot keypoint detection (FSKD) and patch relation learning guided by saliency. (a) We train and test FSKD model via episodes. Each episode consists of support image I^{s}, support keypoints X^{s}, and the query image I^{q}. The goal is to infer the keypoints X in I^{q} that is corresponding to X^{s}. P_{FG} and P_{BG} show the interfering foreground and background patches. Patch $P(X, \Sigma)$ shows the predicted keypoint X with uncertainty Σ by our model (*best-viewed zoom-in*); (b) The saliency map is used to indicate the foreground patches. As a result, our saliency-guided vision transformer (SalViT) is able to enhance the relation learning for the pairwise salient patches (or "tokens") while weaken those of un-salient pairs.

mative keypoint representations for similarity learning, and effectively improve FSKD performance. Our contributions are summarized as follows:

i. We propose a novel masked self-attention (M-SA) by leveraging saliency map as a mask. Our M-SA captures the distant relation between foreground patches and performs soft masking according to the interaction matrix capturing saliency of pairs of tokens.

ii. The saliency map, essentially, defines the *receptive field* of M-SA. Thus, we propose a morphology learner (ML) which automatically adjusts the morphology of saliency map via the so-called power normalization (PN), providing dynamic receptive field for M-SA to exploit useful information in the proximity of foregrounds.

iii. With the M-SA and ML, we present Saliency-guided Vision Transformer (SalViT), and investigate modelling both local and global dependency by fusing CNN and SalViT features.

iv. We propose few-shot keypoint detection (FSKD) with SalViT, and show that SalViT benefits FSKD. We propose a transductive extension of FSKD in addition to inductive FSKD.

2 Related Work

Few-shot Learning (FSL): Family of deep metric space FSL encourage the inter-class distance to be larger while narrowing the intra-class distance. Koch et al. (2015) use deep Siamese networks to model deep metric space. Vinyals et al. (2016) formalize N-way K-shot protocol using episodes. Snell et al. (2017) propose Prototypical networks which forms centroid for each class. Dynamic weight generator methods (Gidaris & Komodakis, 2018; Qiao et al., 2018) use statistics of support features and the weight vectors of base classes to construct classifier. Moreover, Sung et al. (2018) propose a relation module to replace the cosine distance. Another stream of works emphasize on faster adaptation and optimization, *e.g.* MAML (Finn et al., 2017). Finally, data hallucination approaches (Zhang et al., 2019; Zhang & Wang, 2021) increase the data variability in FSL.

Vision Transformers: Transformers (Vaswani et al., 2017) are successful in sequential modeling and natural language processing (*e.g.* BERT (Devlin et al., 2018) and GPT-3 (Brown et al., 2020)). Thus, transformers have also been adopted to vision problems (Parmar et al., 2018). For example, Vision Transformer (ViT) (Dosovitskiy et al., 2021) has inspired a series of advances such as training data efficient DeiT (Touvron et al., 2021), self-supervised DINO (Caron et al., 2021), CrossViT (Chen et al., 2021), *etc*. Although above ViTs outperform CNNs in image classification, they do not provide multi-scale features for dense prediction tasks. Thus, other general-purpose architectures such Swin-T (Liu et al., 2021) and Twins (Chu et al., 2021) were proposed.

Attention in Feature Learning: Attention mechanisms include *self-attention* (SA) and *cross-attention* (CA) (Vaswani et al., 2017; Hou et al., 2019). SA is crucial in ViT modelling but its

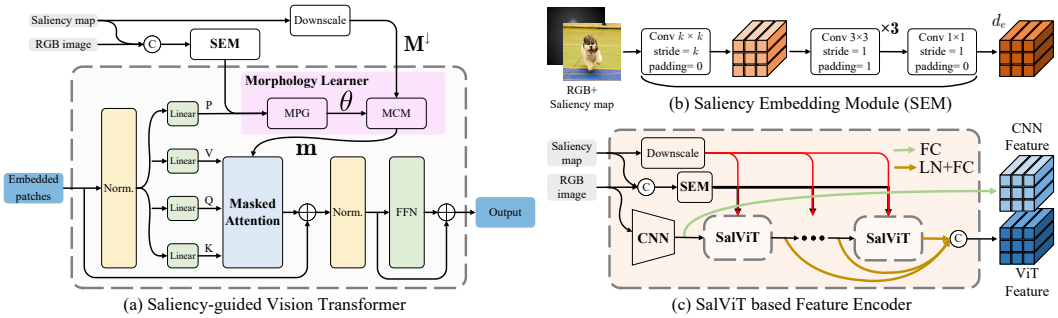


Figure 2: Model overview. (a) The structure of saliency-guided vision transformer (SalViT); (b) details of semantic embedding module (SEM); (c) SalViT based feature encoder $\mathcal{E}$ for FSKD.

computation is high as SA operates globally. Thus, Swin-T (Liu et al., 2021) applies SA within local windows to reduce the computational cost; Twins (Chu et al., 2021) performs SA in each sub-window firstly and then globally. Unlike Swin-T and Twins using regular windows (*i.e.* rectangle) with hard masking, we use saliency map to attain universal foreground shapes of soft masking, realizing the attention within foregrounds. In fact, masked attention is popular and has been used for various purposes, *e.g.* masking invalid tokens, self-supervised training, image inpainting (Li et al., 2022), *etc*. However, our self-attention (M-SA) in our SalViT is uniquely developed for FSKD. Attention has also been explored to complement CNNs, *e.g.* Attention Augmented CNN (Bello et al., 2019) and Squeeze-and-Excitation Networks (Hu et al., 2018). Thus, we attempt to combine ViT and CNN features for FSKD. Moreover, we explore the M-SA with dynamic receptive field.

3 Method

3.1 Preliminary

Notations are in Section A.1. Below we revisit ViT (Dosovitskiy et al., 2021). Given a sequence of n tokens in $\mathbf{X} \in \mathbb{R}^{n\times d}$ with d dimensional feature vectors, $\mathbf{X}$ forms $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$ (*i.e.* query, key, and value)

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_q \in \mathbb{R}^{n\times d}, \quad \mathbf{K} = \mathbf{X}\mathbf{W}_k \in \mathbb{R}^{n\times d}, \quad \mathbf{V} = \mathbf{X}\mathbf{W}_v \in \mathbb{R}^{n\times d}, \tag{1}$$

where $\mathbf{W}_q$, $\mathbf{W}_k$, and $\mathbf{W}_v$ are projection matrices. Then the self-attention (SA) is adopted to learn the relations of all token pairs exhaustively

$$A_{i,j} = \gamma(\phi(\mathbf{Q}_{i,:}, \mathbf{K}_{j,:})), \quad \tilde{\mathbf{V}} = \mathbf{A}\mathbf{V}, \quad \mathbf{z} = \tilde{\mathbf{V}}\mathbf{W}_o, \tag{2}$$

where $\gamma(\cdot)$ is a non-linear function and $\phi(\cdot)$ is a similarity function, the two together yielding attention $\mathbf{A} \in \mathbb{R}^{n\times n}$, $\mathbf{W}_o$ is a projection matrix, and $\mathbf{z}$ is the output of SA. (γ, ϕ) can be the *scaled dot-product based softmax SA*, *i.e.*, $\gamma(\cdot) = \text{softmax}(\cdot)$, $\phi(\mathbf{Q}_{i,:}, \mathbf{K}_{j,:}) = \mathbf{Q}_{i,:}\mathbf{K}_{j,:}^\intercal/(\beta\sqrt{d})$, or *Gaussian kernel based RBF SA*, *i.e.*, $\gamma(\cdot) = \exp(\cdot)$, $\phi(\mathbf{Q}_{i,:}, \mathbf{K}_{j,:}) = -\|\mathbf{Q}_{i,:} - \mathbf{K}_{j,:}\|/(2\beta\sqrt{d})$. To accelerate computation, $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$ are usually split into multiple sub-matrices in channel dimension and computed via multi-head self-attention (MHSA). Finally, a feedforward network (FFN) comprised by two linear layers with GELU activation is used. With the layer normalization (LN) and skip connection, a basic ViT block becomes

$$\mathbf{z}'_l = \text{MHSA}(\text{LN}(\mathbf{z}_{l-1})) + \mathbf{z}_{l-1}, \quad \mathbf{z}_l = \text{FFN}(\text{LN}(\mathbf{z}'_l)) + \mathbf{z}'_l, \tag{3}$$

where $\mathbf{z}_l$ is the output of l-th ViT. Based on vanilla ViT, we will build our SalViT in next section.

3.2 Saliency-guided Vision Transformer

The overview of our SalViT is shown in Figure 2(a), which includes masked self-attention (M-SA), morphology learner (ML), and saliency embedding module (SEM), in addition to the normal LN and FFN in vanilla ViT. The ML takes the patch features $\mathbf{P}$ and the saliency embeddings produced by SEM as input and predicts conditional parameter θ to change the morphology of saliency map, yielding an adapted map $\mathbf{M}$ for M-SA. In what follows, we will detail each component of SalViT.

3.2.1 SALIENCY MAP GENERATION

Visual saliency correlates with human visual attention and is indicated by a single-channel map with values in $[0, 1]$ range. The conventional saliency detectors are modelled by various handcrafted priors, *e.g*. Spetral Residual (SR) (Hou & Zhang, 2007) and Robust Background Detector (RBD) (Zhu et al., 2014), which underperform on complex scenes. The deep saliency detectors include i) fully-supervised detectors, *e.g*. SCRN (Wu et al., 2019) and RFCN (Wang et al., 2016); ii) weakly-supervised ones, *e.g*. MNL (Zhang et al., 2018) and WScrible (Zhang et al., 2020). We use the off-the-shelf saliency detectors to generate saliency maps. As the quality of saliency map may affect SalViT and the FSKD datasets involve complex scenes, we choose SCRN as the default detector. To smooth saliency map and help learn morphology, the maps are pre-processed on-the-fly by efficient distance transform (DT) (Felzenszwalb & Huttenlocher, 2012) and Gaussian blur (Figure 4).

3.2.2 SOFT MASKED SELF-ATTENTION

Recall that the vanilla SA can be formulated as $A_{i,j} = \gamma(\phi(\mathbf{Q}_{i,:}, \mathbf{K}_{j,:}))$ (Eq. 2). Once we have the downscaled saliency map $\mathbf{M}^{\downarrow}$ flattened as $\mathbf{m} \in \mathbb{R}^n$ (n is the number of tokens), we incorporate saliency and extend vanilla SA to $A_{i,j} = \gamma(\psi(\mathbf{Q}_{i,:}, \mathbf{K}_{j,:}, m_i, m_j))$, where m_i, m_j are the saliency of i-th and j-th patches, respectively. Intuitively, the attention is expected to focus on foreground when given $\mathbf{m}$, to capture the patch relations on target object. One may define (γ, ψ) as

$$\gamma(\psi(\mathbf{Q}_{i,:}, \mathbf{K}_{j,:}, \mathcal{S}_+, \mathcal{S}_-)) = \begin{cases} \gamma(\phi(\mathbf{Q}_{i,:}, \mathbf{K}_{j,:})) & i, j \in \mathcal{S}_+ \\ 0 & i \in \mathcal{S}_+, j \in \mathcal{S}_- \\ \delta(i-j) & i \in \mathcal{S}_- \end{cases}, \tag{4}$$

where $\mathcal{S}_+ = \{i : m_i \geq 0.5\}$ (or $\mathcal{S}_- = \{i : m_i < 0.5\}$) stands for the index set of foreground (or background) patches and $\delta(i-j)$ is 1 if $i = j$ otherwise is 0. Our choice of (γ, ψ) means that the relation of the pair of foreground patches is learned, foreground-background pair is inactive, and background patch only focuses self-information. However, such hard masking is inapplicable in real scenario as one patch grid may mix both foreground and background information. Thus, we propose a soft masked self-attention (M-SA). Firstly, we define a so-called saliency interaction matrix (SIM) $\bar{\mathbf{M}} \in \mathbb{R}^{n \times n}$, which represents the saliency degree of paired patches. Specifically, we can use 1) dot-product, *i.e*. $\bar{\mathbf{M}} = \mathbf{m}\mathbf{m}^\intercal$; 2) harmonic mean, *i.e*. $\bar{\mathbf{M}} = 2\mathbf{m}\mathbf{m}^\intercal/(\mathbf{m}\mathbf{1}^\intercal + \mathbf{1}\mathbf{m}^\intercal + \epsilon)$ ($\epsilon \to 0$); and 3) arithmetic mean, *i.e*. $\bar{\mathbf{M}} = (\mathbf{m}\mathbf{1}^\intercal + \mathbf{1}\mathbf{m}^\intercal)/2$, *etc*. We select *harmonic mean* by default as it performed better in our experiments. In SIM, the background patch focusing on itself is not reflected. Thus, a term $\mathbb{I} - \text{Diag}(\mathbf{m})$ is added to build the attention mask $\tilde{\mathbf{M}} \in \mathbb{R}^{n \times n}$ as

$$\tilde{\mathbf{M}} = \bar{\mathbf{M}} + \mathbb{I} - \text{Diag}(\mathbf{m}), \tag{5}$$

where $\text{Diag}(\mathbf{m}) \in \mathbb{R}^{n \times n}$ is the diagonal matrix of $\mathbf{m}$. A property of $\tilde{\mathbf{M}}$ is the diagonal elements are all *unit*. We develop soft M-SA by treating $\tilde{\mathbf{M}}$ as a bias:

$$\mathbf{A} = \gamma(\psi(\mathbf{Q}, \mathbf{K}, \mathbf{M})) = \gamma(\phi(\mathbf{Q}, \mathbf{K}) - (\mathbf{1}\mathbf{1}^\intercal - \tilde{\mathbf{M}})J), \tag{6}$$

where J is a scalar that controls the degree of masking, and (γ, ϕ) is the softmax SA or RBF SA.

Property of M-SA: If the whole image is salient, *i.e*. $\mathbf{m} = \mathbf{1} \in \mathbb{R}^n$, then $\tilde{\mathbf{M}} = \bar{\mathbf{M}} = \mathbf{1}\mathbf{1}^\intercal \in \mathbb{R}^{n \times n}$. As a result, $\gamma(\psi(\mathbf{Q}, \mathbf{K}, \mathbf{M})) = \gamma(\phi(\mathbf{Q}, \mathbf{K}))$ which recovers the vanilla SA. If the whole image is non-salient, *i.e*. $\mathbf{m} = \mathbf{0}$, then $\bar{\mathbf{M}} = \mathbf{0}$, $\tilde{\mathbf{M}} = \mathbb{I}$ then $\gamma(\psi(\mathbf{Q}, \mathbf{K}, \mathbf{M})) = \gamma(\phi(\mathbf{Q}, \mathbf{K}) - (\mathbf{1}\mathbf{1}^\intercal - \mathbb{I})J)$ which means that all attention entries are masked except for the diagonal (M-SA learns no relations).

Position Encoding (PE): PE (Vaswani et al., 2017) augments ViT features with the spatial information and breaks the *permutation equivariance*. There are mainly two types of PE: i) Absolute Position Encoding (APE) (Vaswani et al., 2017; Dosovitskiy et al., 2021) and ii) Relative Position Encoding (RPE) (Bello et al., 2019; Liu et al., 2021). In order to maintain translation equivariance and incorporate position relations, following Bello et al. (2019), we adopt RPE in our M-SA. When computing $\phi(\mathbf{Q}_{i,:}, \mathbf{K}_{j,:})$, the key $\mathbf{K}$ will be preferentially added by RPE as $\mathbf{K}_{j,:} \leftarrow (\mathbf{K}_{j,:} + \mathbf{R}^W_{j_x - i_x,:} + \mathbf{R}^H_{j_y - i_y,:})$, where $j_x - i_x$ and $j_y - i_y$ are relative width and height used to retrieve representations from learnable RPE matrice $\mathbf{R}^W \in \mathbb{R}^{(2W-1) \times d}$ and $\mathbf{R}^H \in \mathbb{R}^{(2H-1) \times d}$.

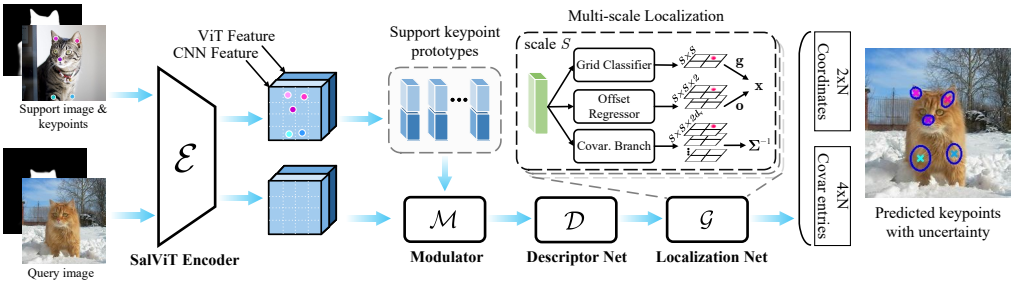


Figure 3: Our few-shot keypoint detection pipeline with SalViT.

3.2.3 Morphology Learner

Although M-SA can directly use the downscaled saliency map $\mathbf{M}^{\downarrow}$, it lacks adaptation to the context of object. To endow SalViT with a dynamic receptive field, we propose the morphology learner (ML) which modifies $\mathbf{M}^{\downarrow}$. As shown in Figure 2(a), the morphology learner consists of a morphology parameter generator (MPG) and a morphology changing module (MCM). Firstly, we adopt a saliency embedding module (SEM) to encode saliency map and RGB image into saliency embeddings $\mathbf{F}_{\text{sal}} \in \mathbb{R}^{n \times d_e}$ (Figure 2(b)). Then, the patch features $\mathbf{P}$ and $\mathbf{F}_{sal}$ are concatenated and globally average pooled as $\mathbf{F} = \text{GAP}([\mathbf{F}_{\text{sal}}, \mathbf{P}]) \in \mathbb{R}^{(d+d_e)}$ and fed to MPG for predicting conditional scalar parameter θ

$$\theta = \mathbf{W}_2 \texttt{GeLU}(\mathbf{W}_1 \mathbf{F} + \mathbf{b}_1) + \mathbf{b}_2. \tag{7}$$

Finally, θ is injected into MCM to control the morphology of $M^{\downarrow}$ via *power normalization* as

$$\tilde{\theta} = \rho_1 \cdot \text{sigmoid}(\theta), \quad \mathbf{M} = (\mathbf{M}^{\downarrow})^{\tilde{\theta}}, \tag{8}$$

where $\rho_1 \in \mathbb{R}$ controls the bound of morphology ($0 \leq \tilde{\theta} \leq \rho_1$). We further add a regularization loss $\mathcal{L}_{\text{reg}} = \mathbb{E}[\max((\tilde{\theta} - \rho_2)^2 - \rho_3, 0)]$, where ρ_2 is the desired morphology center and ρ_3 is interval. We set $\rho_2 = 0.7$ and $\rho_3 = 0.05$. Thus, matrix $\mathbf{M}$ (or $\mathbf{m}$ in vectorized form) is learnable and helps M-SA focus on foregrounds and the context (Figure 1(b)).

3.3 Few-shot Keypoint Detection

Figure 3 shows our few-shot keypoint detection (FSKD) pipeline with SalViT. For K support images and N support keypoints per image, the goal of FSKD is to detect the corresponding keypoints in a query image (*N-way-K-shot detection*). Our FSKD model includes SalViT-based feature encoder $\mathcal{E}$, feature modulator $\mathcal{M}$, descriptor net $\mathcal{D}$, and keypoint localization net $\mathcal{G}$. Features of $\mathcal{E}$ capture relations between salient patches, helping modulator $\mathcal{M}$.

Feature Augmentation with SalViT: Figure 2(c) shows the architecture of $\mathcal{E}$, which takes both saliency and RGB images $\mathbf{I} = [\mathbf{I}^{\text{s}}; \mathbf{I}^{\text{rgb}}]$ as input and outputs ensembled CNN and ViT features as $\mathcal{E}(\mathbf{I}) = [\mathbf{F}^{\text{cnn}}; \mathbf{F}^{\text{vit}}] \in \mathbb{R}^{l \times l \times d}$ where $d = d_{\text{cnn}} + d_{\text{vit}}$. In this way, $\mathcal{E}$ captures both local and global features that aid the keypoint inference. The details of $\mathcal{E}$ is at Section A.3.

Inductive Inference: Assume the k-th support image with saliency $\mathbf{I}^{\text{s}}_k$ are encoded as $\mathcal{E}(\mathbf{I}^{\text{s}}_k) \in \mathbb{R}^{l \times l \times d}$, and $\mathbf{X}^{\text{s}}_{k,n}$ is the n-th support keypoint in k-th support image. Firstly, we extract support keypoint representation (SKR) $\mathbf{\Phi}_{k,n} \in \mathbb{R}^d$ via Gaussian pooling, *i.e.* $\mathbf{\Phi}_{k,n} = \sum_{\mathbf{p}} \exp(- \parallel \mathbf{p} - \mathbf{X}^{\text{s}}_{k,n} \parallel^2_2 / 2\xi^2) \cdot \mathcal{E}(\mathbf{I}^{\text{s}}_k)(\mathbf{p})$, where $\mathcal{E}(\mathbf{I}^{\text{s}}_k)(\mathbf{p})$ means the feature vector at position $\mathbf{p} \in \mathbb{R}^2$. Following Prototypical networks (Snell et al., 2017), we form the support keypoint prototype (SKP) to represent each type of keypoints by averaging the SKRs across support images, namely $\mathbf{c}_n = \frac{1}{K} \sum_{k=1}^{K} \mathbf{\Phi}_{k,n}$. As a result, we form N SKPs in total for N-way-K-shot detection. Subsequently, the SKP are used to modulate query feature map $\mathcal{E}(\mathbf{I}^{\text{q}}) \in \mathbb{R}^{l \times l \times d}$ via $\mathcal{M}$, yielding the attentive feature $\mathbf{F}^n_{\text{att}} = \mathcal{M}(\mathcal{E}(\mathbf{I}^{\text{q}}), \mathbf{c}_n) \in \mathbb{R}^{l \times l \times d}$ per SKP, where $\mathbf{F}^n_{\text{att}}(\mathbf{p}) = \mathcal{E}(\mathbf{I}^{\text{q}})(\mathbf{p}) \odot \mathbf{c}_n$. $\mathcal{M}$ correlates SKPs with similar regions in the query image. Each $\mathbf{F}^n_{\text{att}}$ is processed into descriptor $\mathbf{\Psi}_n$ via $\mathcal{D}$ and fed to a multi-scale grid-based localization net $\mathcal{G}$ to predict keypoint $\mathbf{x} \in \mathbb{R}^2$ and uncertainty $\mathbf{\Sigma} \in \mathbb{R}^{2 \times 2}$, as in Fig. 3.

Each scale of $\mathcal{G}^S$ has three branches, *i.e.* grid classifier $\mathcal{G}^S_{\text{c}}$, offset regressor $\mathcal{G}^S_{\text{o}}$, and covariance branch $\mathcal{G}^S_{\text{v}}$, which are used to predict grid probability map $\mathbf{P}^S = \text{softmax}(\mathcal{G}^S_{\text{c}}(\mathbf{\Psi}_n)) \in \mathbb{R}^{S \times S}$, offset

vector field $\mathbf{O}^S = \mathcal{G}_{\text{o}}^S(\mathbf{\Psi}_n) \in \mathbb{R}^{S\times S\times 2}$, and latent covariance field $\mathbf{V}^S = \mathcal{G}_{\text{v}}^S(\mathbf{\Psi}_n)) \in \mathbb{R}^{S\times S\times 2d_{\text{v}}}$, respectively. The grid $\mathbf{g}^S \in \{0,\cdots,S\text{–}1\}\times\{0,\cdots,S\text{–}1\}$ that has maximal score is indexed from $\mathbf{P}^S$. Accordingly, we extract offset $\mathbf{o}^S = O^S(\mathbf{g}^S) \in [-1,1)^2$ and latent covariance $\mathbf{Q}^S = \mathbf{V}^S(\mathbf{g}^S) \in \mathbb{R}^{2d_v}$ using $\mathbf{g}^S$. In this paper, we regress precision matrix $\mathbf{\Omega}^S = \tilde{\mathbf{Q}}^S(\tilde{\mathbf{Q}}^S)^\intercal/d_v$ instead of covariance $\mathbf{\Sigma}^S$ to avoid computing matrix inverse in loss function, where $\tilde{\mathbf{Q}}^S \in \mathbb{R}^{2\times d_v}$ is reshaped from $\mathbf{Q}^S$. As such, one may write the grid classification loss $\mathcal{L}_{\text{cls}}^S$ and offset regression loss $\mathcal{L}_{\text{os}}^S$ at scale S as

$$\mathcal{L}_{\text{cls}}^S = -\mathbb{E}\log \mathbf{P}^S(\hat{\mathbf{g}}), \quad \mathcal{L}_{\text{os}}^S = \frac{1}{2}\mathbb{E}[(\mathbf{x}^S - \hat{\mathbf{x}})^\intercal \mathbf{\Omega}^S(\mathbf{x}^S - \hat{\mathbf{x}}) - \log\det(\mathbf{\Omega}^S)], \tag{9}$$

where $\hat{\mathbf{g}}$ and $\hat{\mathbf{x}}$ are groundtruth grid and offset. If $\mathcal{G}$ has N_S scales $\{S_i\}_{i=1}^{N_S}$, the multi-scale localization loss is $\mathcal{L}_{\text{ms}} = \sum_{i=1}^{N_S}(\mathcal{L}_{\text{cls}}^{S_i} + \mathcal{L}_{\text{os}}^{S_i})/N_S$. Moreover, the multi-scale prediction is given as

$$\mathbf{x} = \frac{1}{N_S}\sum\nolimits_{i=1}^{N_S}\frac{l_0}{S_i}(\mathbf{g}^{S_i} + 0.5 + 0.5\mathbf{o}^{S_i}), \quad \mathbf{\Sigma} = \frac{1}{4N_S}\sum\nolimits_{i=1}^{N_S}\left(\frac{l_0}{S_i}\right)^2\mathbf{\Sigma}^{S_i}, \tag{10}$$

where l_0 is the edge length of square-padded query image, $\mathbf{\Sigma}^{S_i}$ is the inverse of $\mathbf{\Omega}^{S_i}$. Considering the regularization loss $\mathcal{L}_{reg}$ imposed from $\mathcal{E}$, our overall loss becomes

$$\mathcal{L} = \lambda_1\mathcal{L}_{\text{ms}} + \lambda_2\mathcal{L}_{\text{reg}} \qquad (\text{we use } \lambda_1 = \lambda_2 = 0.5). \tag{11}$$

Transductive Inference: A key issue of FSKD is the need to represent prototype $\mathbf{c}_n$ for each kind of support keypoints well, and measure its similarity w.r.t. query. However, as support samples are limited, SKP $\mathbf{c}_n$ do not represent the true distribution of a given keypoint type well. Thus, we extend FSKD to the transductive setting by leveraging the unlabeled query samples.

Let SKRs be $\mathcal{S} = \{\mathbf{\Phi}_{k,n}\}_{k,n=1}^{K\times N}$ and the SKRs of n-th keypoint type be $\mathcal{S}^{(n)} = \{\mathbf{\Phi}_{k,n}\}_{k=1}^{K}$. Let the initial SKPs $\mathcal{C} = \{\mathbf{c}_n\}_{n=1}^{N}$ be built based on $\mathcal{S}$. For each query image and each prototype, we can retrieve top-W query keypoints. For Z query images, we get $\{\mathbf{x}_{z,n,w}\}_{z,n,w=1}^{Z\times N\times W}$, by selecting the top-$W$ grids from the grid probability map. Then we define the label confidence criterion by using the associate keypoint probability scores $\{p_{z,n,w}\}_{z,n,w=1}^{Z\times N\times W}$, rank the scores per keypoint type in descending order, and select top-η query keypoints per type to generate the candidate set $\mathcal{Q}^{(n)} = \{\tilde{\mathbf{\Phi}} : \tilde{\mathbf{\Phi}} = \mathcal{E}(\mathbf{I}_z^{\text{q}})(\mathbf{x}_{z,n,w}), S_{z,w}^{(n)} = 1,\ \forall z,w\}$, where $\mathcal{E}(\mathbf{I}_z^{\text{q}})(\mathbf{x}_{z,n,w})$ is the keypoint representation extracted at $\mathbf{x}_{z,n,w}$ from the query image $\mathbf{I}_z^{\text{q}}$, $\mathbf{S}^{(n)} \in \{0,1\}^{Z\times W}$ denotes the selection indicator for n-th type of predicted query keypoints. We solve

$$\mathbf{S}^{(n)} = \arg\max\nolimits_{\|\mathbf{S}'^{(n)}\|_0=\eta}\sum\nolimits_{z,w=1}^{Z\times W} p_{z,n,w}S'^{(n)}_{z,w}. \tag{12}$$

Based on the candidate set $\mathcal{Q}^{(n)}$, we refine SKPs using the soft assignment as

$$\mathbf{c}_n^* = \frac{1}{\kappa|\mathcal{S}^{(n)}| + (1-\kappa)\sum_{\tilde{\mathbf{\Phi}}\in\mathcal{Q}^{(n)}} p(\tilde{\mathbf{\Phi}}, \mathbf{c}_n)}\left(\kappa\sum\nolimits_{\mathbf{\Phi}\in\mathcal{S}^{(n)}}\mathbf{\Phi} + (1-\kappa)\sum\nolimits_{\tilde{\mathbf{\Phi}}\in\mathcal{Q}^{(n)}} p(\tilde{\mathbf{\Phi}}, \mathbf{c}_n)\tilde{\mathbf{\Phi}}\right), \tag{13}$$

where $0 \le \kappa \le 1$ controls the amount of pseudo-label information added, and $p(\tilde{\mathbf{\Phi}}, \mathbf{c}_n) = \exp(-\|\tilde{\mathbf{\Phi}} - \mathbf{c}_n\|/2\sigma^2)/\sum_{i=1}^{N}\exp(-\|\tilde{\mathbf{\Phi}} - \mathbf{c}_i\|/2\sigma^2)$ is the assignment probability given $\mathbf{c}_n$. The updated SKP $\mathbf{c}_n^*$ is more representative and utilized to re-estimate query keypoints.

4 Experiments

4.1 Settings

Datasets: We conduct FSKD experiments on *five* datasets. *Animal pose* dataset (Cao et al., 2019) has five mammal species *cat*, *dog*, *cow*, *horse*, and *sheep*, with 6000+ instances with keypoint annotations. *CUB* (Wah et al., 2011) has 200 species, 15 keypoint annotations, and 11788 images. *NABird* (Van Horn et al., 2015) is more complex and spans 555 categories, 11 keypoint annotations, and 48562 images. *AwA* (Banik et al., 2021) has 35 animal species with 10064 images. *DeepFashion2* (Ge et al., 2019) is a large-scale clothing dataset with 13 categories and 8∼39 landmarks.

Metric: The percentage of correct keypoints (PCK) (Yang & Ramanan, 2012) is used, where a correct keypoint prediction requires its distance to groundtruth (GT) less than $\tau\cdot\max(w_{\text{bbx}}, h_{\text{bbx}})$, where w_{bbx} and h_{bbx} are the edges of object bounding box. Generally, τ is set to 0.1.

Table 1: Results on 1-shot/5-shot keypoint detection for unseen species across four datasets. PCK metric is used and the 95% confidence intervals of all scores are below 1.2%. The average of five subproblems in Animal pose dataset is reported. The results marked with * are copied from original paper. ** and **+T** refer to apply finetuning or transductive inference during testing, respectively. The best and second best results are marked in **bold** and underline.

Setting	Model	Animal Pose		CUB		NABird		AwA	
		1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
Novel	ProbIntr (Novotny et al., 2018)	21.25	34.31	68.07	75.65	48.70	60.86	32.00	51.68
	TFA** (Wang et al., 2020a)	20.82	41.39	50.12	73.43	32.65	57.77	28.68	58.64
	ProtoNet (Snell et al., 2017)	15.47	23.66	51.32	61.17	36.65	52.56	29.57	36.94
	RelationNet (Sung et al., 2018)	16.99	21.37	56.59	60.41	34.02	41.42	20.91	28.58
	WG (w/o Att.) (Gidaris & Komodakis, 2018)	17.53	21.92	52.66	61.77	33.31	47.78	28.13	35.23
	WG (Gidaris & Komodakis, 2018)	18.40	27.70	54.75	60.70	34.19	48.03	28.54	40.02
	FSKD (rand)* (Lu & Koniusz, 2022)	38.77	51.42	77.90	80.26	54.01	63.76	51.81	68.39
	SalViT (rand)	41.86	51.98	**79.47**	**81.90**	56.30	**64.88**	54.56	70.11
	SalViT (rand)+T	42.44	52.08	79.24	81.63	56.35	64.55	56.06	70.55
	FSKD (default)* (Lu & Koniusz, 2022)	44.75	54.63	77.89	79.17	56.04	63.46	64.76	74.47
	SalViT (default)	46.93	55.77	78.76	79.85	**56.41**	64.49	66.05	76.40
	SalViT (default)+T	**47.75**	**56.23**	78.42	79.45	56.08	64.22	**66.77**	**76.71**
Base	ProbIntr (Novotny et al., 2018)	40.78	39.29	73.46	82.18	70.56	79.01	55.72	54.88
	ProtoNet (Snell et al., 2017)	37.73	44.94	80.36	86.92	73.18	83.67	57.17	62.49
	RelationNet (Sung et al., 2018)	43.18	47.91	79.40	84.43	78.85	84.46	57.31	66.98
	WG (w/o Att.) (Gidaris & Komodakis, 2018)	43.69	45.69	80.42	87.80	80.83	83.21	62.77	65.85
	WG (Gidaris & Komodakis, 2018)	43.87	49.83	81.96	88.33	81.39	83.49	63.87	69.81
	FSKD (rand)* (Lu & Koniusz, 2022)	49.90	55.42	87.94	94.95	87.84	93.03	65.26	75.11
	SalViT (rand)	51.27	56.79	89.42	95.10	88.48	93.96	66.67	76.85
	SalViT (rand)+T	53.22	57.65	90.33	95.24	**89.27**	**94.16**	69.14	**77.55**
	FSKD (default)* (Lu & Koniusz, 2022)	49.93	56.44	87.71	94.53	86.99	93.18	66.39	75.22
	SalViT (default)	52.30	57.24	90.54	95.16	87.95	93.63	68.07	76.65
	SalViT (default)+T	**53.94**	**57.99**	**91.05**	**95.41**	88.94	93.77	**69.43**	77.09

Table 2: 1-shot/5-shot novel keypoint detection on DeepFashion2. The metric of NE (normalized error $d' = d/\max(w_{\text{img}}, h_{\text{img}})$) and PCK with 95% confidence intervals are used. MetaCloth (Ge et al., 2021) tests in 700 episodes and uses 12 gradient steps of finetuning. Our model tests in 1000 episodes with 10 gradient steps of finetuning. The results marked with * are from (Ge et al., 2021).

Method	1-shot		5-shot	
	PCK (↑)	NE (↓)	PCK (↑)	NE (↓)
MAML* (Finn et al., 2017)	-	0.215 ± 0.023	-	0.126 ± 0.017
ProtoNet* (Snell et al., 2017)	-	0.179 ± 0.042	-	0.135 ± 0.039
WG* (Gidaris & Komodakis, 2018)	-	0.185 ± 0.051	-	0.160 ± 0.047
FSKD (rand) (Lu & Koniusz, 2022)	33.04 ± 1.80	0.149 ± 0.005	44.92 ± 1.72	0.111 ± 0.004
MetaCloth* (Ge et al., 2021)	-	0.145 ± 0.021	-	0.096 ± 0.015
SalViT (rand) (Ours)	$\mathbf{35.01 \pm 1.47}$	$\mathbf{0.130 \pm 0.004}$	$\mathbf{50.87 \pm 1.18}$	$\mathbf{0.093 \pm 0.002}$

Compared Methods: We adapt *ProbIntr* proposed by (Novotny et al., 2018) for FSKD comparison. For fairness, we train ProbIntr jointly by self-supervised keypoints and labeled query keypoints. Then, popular few-shot learning approaches *ProtoNet* (Snell et al., 2017), *RelationNet* (Sung et al., 2018), *WG* (with or w/o attention) (Gidaris & Komodakis, 2018), and Two-stage Finetuning Approach (*TFA*) (Wang et al., 2020a) are adapted to FSKD. Moreover, two models, *FSKD (rand)* and *FSKD (default)* (Lu & Koniusz, 2022) are used. They use two kinds of auxiliary keypoints in training (*i.e.*, *random* or *default* interpolated keypoints, see Section B.3). To improve keypoint generalization, we also apply two auxiliary keypoint variants *SalViT (rand)* and *SalViT (default)*.

Implementation Details: The RGB image and saliency map use simultaneous transformations including random flips, rotations, and crops. The image is resized and padded into 384×384 as input to FSKD model. The CNN module inside SalViT-based feature encoder $\mathcal{E}$ is modified from ResNet50 (He et al., 2016) and yields $12 \times 12 \times 2048$ patch embeddings $\mathbf{F}^{\text{cnn}}$. By default, we use one SalViT with RBF-based M-SA, l_2 normalize the queries and keys, set the temperature $\beta = 1$ and the degree of masking $J = 1$. The final output $\mathbf{F}^{\text{vit}}$ has 768 channels by default (*i.e.* $d_{\text{vit}}/d_{\text{CNN}} = 0.375$). All models are trained with 40k episodes and the scores are averaged over 1000 episodes in testing. For transductive inference, we set $W = 2$, $\sigma = 0.05$, and $\kappa = 0.8$.

4.2 COMPARISONS ON FEW-SHOT KEYPOINT DETECTION

The FSKD models are trained on seen species with base keypoints, and tested on unseen species with novel keypoints. Same as (Lu & Koniusz, 2022), the novel keypoints are selected as follows: *two eyes* and *four knees* for Animal and AwA pose dataset; *forehead*, *two eyes*, and *two wings* for CUB; *two eyes* and *two wings* for NABird. The remaining keypoints are used by the base keypoint set. Section B.2 shows the full splits. For Animal pose dataset, we alternately choose one animal kind as unseen species for testing while the remaining four as seen species for training, which yields five subproblems. For CUB, 100 species are for training, 50 for validation, and 50 for testing. For NABird, the split is 333, 111, and 111 for training, validation and testing respectively. In terms of AwA, 25 species are for training and 10 for testing. For DeepFashion2, 6 upper-body clothing categories are set as seen categories and 7 lower-body clothing categories are set as unseen ones.

Table 1 shows the 1-shot/5-shot results on four datasets. Our model variants obtain best PCK scores on all tasks and enjoy significant improvements on both novel and base keypoint detection against FSKD of (Lu & Koniusz, 2022), achieving 41.86%, 79.47%, 56.30%, 54.56% of 1-shot novel keypoint detection scores in Animal, CUB, NABird, and AwA datasets for SalViT (rand) and 46.93%, 78.76%, 56.41%, 66.05% for SalViT (default). For transductive inference, most of results further increase, which indicates the unlabeled data enhances keypoint representations. As DeepFashion2 has a varying number of keypoints in each clothing category, we adopt the SalViT (rand) which uses random auxiliary keypoints to perform 1/5-shot novel keypoint detection. Table 2 shows our SalViT (rand) achieves higher PCK scores yet lower NE error than other methods.

4.3 ABLATION STUDY

Ablatiosn are performed on SalViT (default) under the setting of novel keypoint detection.

CNN *vs*. ViT: CNN-only (relies on neighborhood information) or vanilla ViT-only (relies on global patch relations) based models are our baselines. Table 3 shows both models have similar FSKD scores due to their own merits. When combining both CNN and ViT features, the model shows either best or second best results compared to individual CNN-only and ViT-only model.

M-SA, Morphology Learner (ML) and PE: When replacing SA in vanilla ViT with our M-SA using raw saliency map (without morphology changing) as mask, the scores have up 2% gain across four datasets (4-th row, Table 3), which validates the importance of foreground relations in FSKD. When adding M-SA together with ML, the results further improve (8-th row, Table 3). Moreover, we investigate the impacts of various position encodings (PE) in FSKD. By comparing 'not using PE', 'using absolute PE' (APE), and RPE, the model with RPE yields best scores (relative spatial information helps infer keypoints). We notice that the RPE used in our model yields better scores than RPE* from SwinT, which may be due to that our adopted RPE is a learnable vector for each relative position, containing richer information than RPE* which is scalar. Moreover, we perform experiments by manually setting power normalization (PN) value θ against using ML to automatically learn PN value conditioned on images. Table 4 shows that proper morphology changing yields best performance (ML is better than a manually set parameter).

Saliency Detectors: Table 5 shows the SCRN saliency detector yields best performance. However, the weakly-supervised MNL narrows the gap with SCRN when increasing support images to 5 shots.

4.4 FSKD UNDER OCCLUSION

Below we investigate if our masked self-attention (M-SA) is robust to keypoint occlusions. We consider various strategies to improve the occlusion handling ability of our model.

Firstly, we test the normally-trained FSKD model under different occlusion probability to query keypoints. The occlusion are *gray box*, *average-pixel box* (a box filled by average pixel), or *background crop*. Each testing occlusion block is centered at query keypoint and has random size with an area ratio to object bounding box $0.01 \sim 0.04$ and aspect ratio $0.7 \sim 1.4$. Secondly, we explore masking different signals during training: RGB image, saliency map, and CNN patch features. The $2 \sim 12$ foreground patches in RGB image are randomly masked with gray box, saliency and CNN features are masked with zeros. The FSKD model trained by various masking strategies and tested under occluded query keypoints. Table 6 shows that when no masking is applied, the normally-trained

Table 3: Ablation study of SalViT in 1-shot novel keypoint detection. M-SA: masked self-attention; PE: position encoding; APE*: learnable absolute PE (Dosovitskiy et al., 2021); RPE*: relative PE same to SwinT (Liu et al., 2021) but different to ours. The results on Animal dataset is the average of five subproblems.

Models	Animal	CUB	NABird	AwA
1: *FSKD+CNN-only*	44.25	76.54	54.71	61.75
2: *FSKD+ViT-only*	43.19	76.71	54.44	62.53
3: *FSKD+CNN+ViT*	44.34	76.62	54.57	63.22
4: *FSKD+SalViT (w. M-SA)*	45.30	78.11	55.16	64.48
5: *FSKD+SalViT (w/o PE)*	44.84	76.26	55.41	64.04
6: *FSKD+SalViT (w. APE*)*	44.34	76.82	54.82	62.73
7: *FSKD+SalViT (w. RPE*)*	45.81	78.41	55.97	65.11
8: *FSKD+SalViT*	**46.93**	**78.76**	**56.41**	**66.05**

Table 4: Influence of power normalization (PN) value to SalViT in FSKD. The *auto* means using learnable PN, namely turning on the Morphology Learner in SalViT. The saliency map is pre-processed by image processing to ensure smoothness for PN. The best and second best results are marked in **bold** and underline.

Datasets	0.2	0.5	0.7	0.8	1.0	2.0	5.0	auto
Cat	54.05	54.80	**55.36**	53.88	53.15	53.48	53.92	55.20
Dog	45.74	47.11	47.74	47.35	48.03	46.56	46.35	**49.25**
Cow	41.83	42.07	43.45	43.27	43.01	42.73	41.82	**45.83**
Horse	40.65	41.14	42.54	41.99	43.44	43.03	42.39	**44.82**
Sheep	35.55	35.58	37.53	36.67	36.40	35.01	35.58	**39.54**
Avg	43.56	44.14	45.32	44.63	44.81	44.16	44.01	**46.93**
CUB	76.03	77.69	79.97	**80.99**	78.22	77.08	75.24	78.76
NABird	54.99	54.94	56.01	55.99	55.38	54.21	51.10	**56.41**
AwA	61.53	61.57	63.58	62.94	62.23	61.29	60.07	**66.05**

Table 5: Comparison of different saliency detectors. The setting is 1-shot/5-shot keypoint detection in Animal pose dataset and the average score of five subproblems is reported.

Setting	Shots	SR	RBD	WScribble	MNL	SCRN
Novel	1-shot	44.84	43.82	44.03	44.33	**46.93**
	5-shot	54.89	54.38	54.86	55.16	**55.77**
Base	1-shot	50.83	49.95	50.69	50.42	**52.30**
	5-shot	56.56	55.97	56.69	56.92	**57.24**

Table 6: SalViT-based FSKD models trained by various masking strategies under occlusions. ✓ indicates the foreground patch masking is applied over a kind of information (*i.e.* RGB image, saliency map, or CNN patch features); ✗ means no masking. Each row shows a configuration of masking strategy to train FSKD model. The model is tested under various occlusion probability p of query keypoints. The higher p, the more difficult to predict keypoints. The average PCK scores of *five* subproblems in Animal dataset are reported. The occlusion uses **background crop** in testing.

RGB	Saliency	Patch feature	0%	10%	30%	50%	70%	100%
✗	✗	✗	44.60	42.64	38.49	34.05	29.68	22.82
✓	✓	✗	45.28	43.72	40.68	37.70	34.97	29.77
✗	✓	✓	45.18	43.02	38.84	34.63	30.57	23.74
✓	✓	✓	45.04	43.62	40.75	37.42	34.49	29.13
✗	✓	✗	44.43	42.96	38.48	34.30	29.85	22.63
✓	✗	✗	44.60	43.11	40.49	37.94	34.85	30.27

FSKD model shows weak occlusion handling ability. Masking RGB image, RGB + Saliency, RGB + Saliency + CNN patch features show higher yet similar results towards occlusions of *background crop* even though trained with occlusions of *gray box*. More test results on *gray box* and *average-pixel box* occlusions are in Section D.

5 Conclusion

We have introduced a saliency-guided vision transformer, SalViT, for few-shot keypoint detection. The key insight is to learn the foreground patch relations via SalViT and use such relation for similarity learning in FSKD. The SalViT includes a uniquely designed masked self-attention (M-SA) and a morphology learner (ML), which together constrain self-attention within salient regions while allowing adaptation. Moreover, we have presented a transductive FSKD variant in addition to the standard inductive inference. The FSKD benchmarks across five datasets highlight that our FSKD with SalViT achieves competitive results. The extensive experiments show the effectiveness of M-SA and ML, and the occlusion setting reveals that our SalViT-based model can handle occlusions of keypoints.

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A Details on Approach

A.1 Notations

Capitalised bold symbols denote matrix, lowercase bold symbols denote vectors, and regular fonts denote scalars. We define $\mathbf{1}$ as 'all ones' matrix or vector, $\mathbb{I}$ as identity matrix, and $A_{i,j}$ as the (i,j)-th entry of $\mathbf{A}$. Operator : extracts i-th vector $\mathbf{X}_{i,:}$ from matrix $\mathbf{X}$. Diag$(\cdot)$ embeds a vector to form a diagonal matrix.

A.2 Saliency Maps



(a) Origin image (b) RBD (c) WScribble (d) SCRN (e) DT&Blur

Figure 4: Example of saliency maps generated by different detectors. The rightmost shows the preprocessed map of image (d), where the colorized version is adjunctive for better visualizing saliency diffusion from foreground to background.

Some examples of saliency maps are shown in Figure 4.

A.3 ARCHITECTURE DETAILS

Saliency Embedding Module (SEM): Concretely, SEM includes a large-strided convolution (conv.), three 3×3 small-strided conv., and a 1×1 conv., as shown in Figure 2(b).

SalViT based Feature Encoder $\mathcal{E}$: As shown in Figure 2(c), specifically, $\mathcal{E}$ is comprised of a CNN module, one or multiple cascaded SalViTs, and a peripheral sharable SEM, where CNN module is used for extracting preliminary local embeddings $\mathbf{F}_{\text{CNN}}$, while the SalViTs are for learning relations among these local embeddings with a lean towards foreground patches under the guidance of saliency information provided by SEM. We pass the output of each SalViT through LN and fully-connected layer (FC) and then concatenate them into $\mathbf{F}_{\text{ViT}} = [\mathbf{F}^1; \cdots; \mathbf{F}^T]$, where $\mathbf{F}^t \in \mathbb{R}^{l \times l \times (d_{\text{ViT}}/T)}$ and T is the number of SalViTs. Finally, $\mathbf{F}_{\text{CNN}}$ and $\mathbf{F}_{\text{ViT}}$ are combined to form the output $\mathcal{E}(\mathbf{I})$.

Descriptor Net $\mathcal{D}$: The descriptor net contains dedicated conv. layers in order to manipulate their feature maps to desired sizes (*i.e.* $2048 \times 12 \times 12$ for input and $512 \times 2 \times 2$ for output), whereas the intermediate layers contain a series of 3×3 conv. blocks which continuously reduce the feature map size from 12×12 to 2×2. The output feature will be stretched into feature vector as descriptor.

B DETAILS ON FSKD EXPERIMENT SETUP

B.1 IMPLEMENTATION DETAILS

Our Method: The M-SA of our SalViT has 32 attention heads with each head 64 dimension. The localization net $\mathcal{G}$ uses three scales $S = \{8, 12, 16\}$. We use Adam as optimizer and set the learning rate to $1e-4$.

All-way K-shot ($K = 1, 5$) protocol is used on all tasks, namely, in each episode, there are K support images and all base (or novel) keypoints are serve as support keypoints. The episode is formed by same-species images. In addition to detecting novel keypoint on unseen species, we also test performance for base keypoints using identical trained model.

B.2 CONCRETE KEYPOINT SPLITS:

1) **Animal pose dataset.** Base keypoint set: *two ears, nose, four legs, four paws*; Novel keypoint set: *two eyes, four knees*. 2) **AwA pose dataset.** Base keypoint set: *two earbases, nose, neck base, neck end, throat base, throat end, tail, four legs, four paws, belly bottom, belly middle right, belly middle left*; Novel keypoint set: *two eyes, four knees*. 3) **CUB.** Base keypoint set: *beak, belly, back, breast, crown, two legs, nape, throat, tail*; Novel keypoint set: *forehead, two eyes, two wings*. 4) **NABird.** Base keypoint set: *beak, belly, back, breast, crown, nape, tail*; Novel keypoint set: *two eyes, two wings*. 5) **DeepFashion2.** Base keypoint set: *158 total keypoints on 6 upper-body clothing categories*; Novel keypoint set: *136 total keypoints on 7 lower-body clothing categories*. Please check https://github.com/switchablenorms/DeepFashion2 for more details.

B.3 AUXILIARY KEYPOINT INTERPOLATION

Specifically, the auxiliary keypoints are generated via interpolation $\mathcal{T}(t; [\mathbf{u}_1, \mathbf{u}_2])$ on a path whose end points are $[\mathbf{u}_1, \mathbf{u}_2]$, and interpolation node $t \in (0, 1)$. The interpolation path could be constructed by randomly sampling a number of paths formed by arbitrary body parts, or directly using default limbs. These two types of interpolation paths are corresponding two types of auxiliary keypoints, namely *random* version and *default* version. Following Lu & Koniusz (2022), we set $t = 0.25, 0.50, 0.75$ and the number of interpolation path is set to 6.

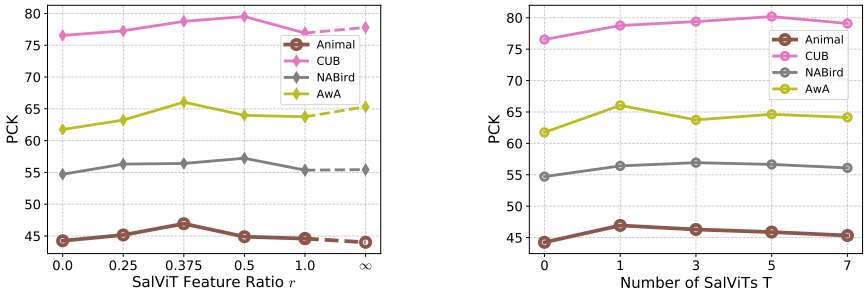
C ADDITIONAL RESULTS ON FSKD

Animal Pose Dataset: Table 7 shows the results of 1-shot/5-shot keypoint detection on each subproblem in Animal pose dataset.

Table 7: Results on 1-shot/5-shot keypoint detection for unseen species of *five* subproblems in Animal pose dataset. PCK metric is used and the 95% confidence intervals of all scores are below 1.2%. The average of five subproblems in Animal pose dataset is reported. The results marked with * are copied from original paper. ** and **+T** refer to finetuning or transductive inference applied during testing, respectively.

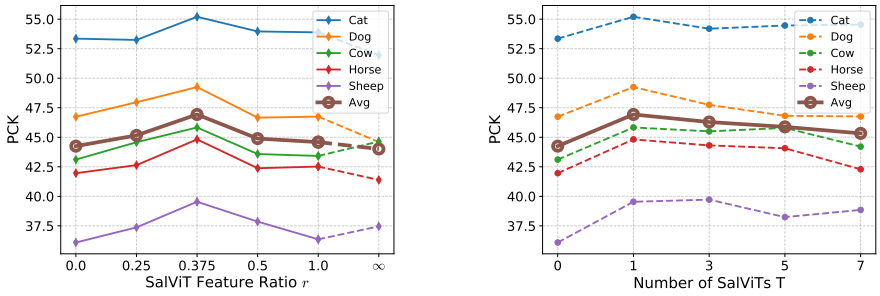
Shots	Setting	Model	Animal Pose Dataset					
			Cat	Dog	Cow	Horse	Sheep	Avg
1-shot	Novel	ProbIntr	28.54	23.20	19.55	17.94	17.03	21.25
		TFA**	22.12	20.00	20.85	20.98	20.14	20.82
		ProtoNet	19.68	16.18	14.39	12.05	15.06	15.47
		RelationNet	22.15	17.19	15.47	13.58	16.55	16.99
		WG (w/o Att.)	21.86	17.11	16.19	16.34	16.13	17.53
		WG	22.47	19.39	16.82	16.40	16.94	18.40
		FSKD (rand)*	46.05	40.66	37.55	38.09	31.50	38.77
		SalViT (rand)	50.41	42.91	40.50	40.74	34.76	41.86
		SalViT (rand)+T	51.13	44.39	40.68	40.80	35.20	42.44
		FSKD (default)*	52.36	47.94	44.07	42.77	36.60	44.75
		SalViT (default)	55.20	49.25	45.83	44.82	39.54	46.93
		SalViT (default)+T	56.07	50.24	46.97	45.35	40.12	47.75
	Base	ProbIntr	45.96	42.49	37.87	40.53	37.04	40.78
		ProtoNet	45.80	39.83	34.88	35.80	32.33	37.73
		RelationNet	51.03	45.85	39.86	41.97	37.19	43.18
		WG (w/o Att.)	51.52	45.50	43.38	40.15	37.89	43.69
		WG	50.05	44.64	43.47	43.35	37.84	43.87
		FSKD (rand)*	57.12	51.12	47.83	49.71	43.71	49.90
		SalViT (rand)	58.76	51.70	50.02	51.30	44.59	51.27
		SalViT (rand)+T	60.49	53.68	52.12	53.04	46.75	53.22
		FSKD (default)*	56.38	51.29	48.24	49.77	43.95	49.93
		SalViT (default)	59.71	51.45	50.98	52.64	46.70	52.30
		SalViT (default)+T	61.53	53.18	52.49	54.29	48.23	53.94
5-shot	Novel	ProbIntr	42.11	38.30	30.58	32.50	28.06	34.31
		TFA**	42.60	38.46	42.40	44.03	39.46	41.39
		ProtoNet	27.41	24.34	22.58	21.78	22.21	23.66
		RelationNet	24.41	20.35	21.81	17.80	22.46	21.37
		WG (w/o Att.)	22.07	20.27	21.83	24.09	21.36	21.92
		WG	31.69	28.12	26.86	27.35	24.48	27.70
		FSKD (rand)*	58.60	51.20	50.95	52.92	43.41	51.42
		SalViT (rand)	58.06	52.73	52.02	52.96	44.12	51.98
		SalViT (rand)+T	58.18	52.93	52.08	53.03	44.17	52.08
		FSKD (default)*	61.78	55.51	55.38	53.21	47.25	54.63
		SalViT (default)	61.19	56.45	56.83	56.79	47.59	55.77
		SalViT (default)+T	61.21	57.07	57.51	57.54	47.84	56.23
	Base	ProbIntr	45.70	42.00	36.81	40.94	31.02	39.29
		ProtoNet	49.30	43.10	45.80	47.04	39.44	44.94
		RelationNet	52.09	48.96	46.47	50.91	41.14	47.91
		WG (w/o Att.)	48.89	45.92	46.16	48.32	39.16	45.69
		WG	54.63	50.57	48.88	51.00	44.06	49.83
		FSKD (rand)*	60.28	54.06	55.51	57.34	49.91	55.42
		SalViT (rand)	61.96	55.76	57.21	58.14	50.89	56.79
		SalViT (rand)+T	62.52	56.70	58.13	59.03	51.89	57.65
		FSKD (default)*	60.89	55.73	57.03	57.14	51.42	56.44
		SalViT (default)	61.46	56.01	58.34	58.61	51.76	57.24
		SalViT (default)+T	62.25	56.63	59.22	59.13	52.70	57.99

SalViT Feature: Recall that our SalViT based encoder $\mathcal{E}$ can adjust the ratio of SalViT feature channels against CNN ones (*i.e.* $r = d_{\text{ViT}}/d_{\text{CNN}}$) by fully-connected layer. Thus we explore the impacts of different features to FSKD performance. Firstly, we fix $d_{\text{CNN}} = 2048$ and vary ratio r. Then, we explore the performance of SalViT-only model (no CNN features). Figure 5a shows the results on Animal, CUB, NABird, and AwA datasets. The score boosts up to 5% when properly



(a) (b)

Figure 5: The influence of SalViT feature ratio and number of SalViTs. (a) The effects as a ratio of increasing SalViT feature channel against fixed CNN feature channel of 2048. The ∞ means SalViT-only with 2048 channels. (b) The impacts of different number of SalViTs. The average of five subproblems in Animal pose dataset is drawn (see Figure 6 for each subproblem).



(a) (b)

Figure 6: Varying SalViT feature ratio and number of SalViTs on each Animal subproblem. (a) The effects as a ratio of increasing SalViT feature channel against fixed CNN feature channel of 2048. The ∞ means SalViT-only with 2048 channels. (b) The impacts of different number of SalViTs.

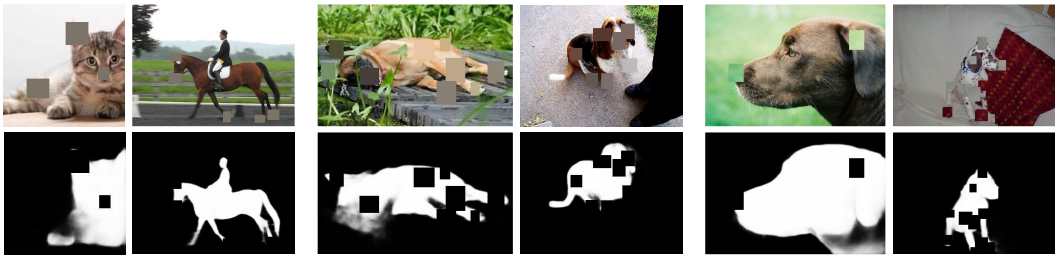
increasing SalViT feature channels, while drops if too much due to overfitting. The best model that uses both CNN and SalViT features also outperforms SalViT-only, which shows right design of $\mathcal{E}$.

Number of SalViTs: Figure 5b shows that our FSKD model is robust to varying number of SalViTs. It yields best performance when cascading one or a few SalViTs while drops if too many. We conjecture that small number of cascaded SalViTs are enough to capture the foreground patch relations within our FSKD datasets.

D Additional Results on FSKD under Occlusion

Occlusions visualization: The occlusions regarding *gray box*, *average-pixel box*, and *background crop* are visualized in Figure 7.

Testing results with occlusions: The testing results of using occlusions of *gray box* are shown in Table 8 and the results of using *average-pixel box* are shown in Table 9. As we can see, the performance gap between no masking and masking RGB + Saliency becomes larger.



(a) Gay box (a) Average-pixel box (a) Background crop

Figure 7: Visualization of three types of occlusions used in testing.

Table 8: The occlusion handling ability of SalViT based FSKD models trained by various masking strategies. ✓indicates the foreground patch masking is applied over a kind of information (*i.e.* RGB image, saliency map, or CNN patch features); ✗means no masking. Each row shows a configuration of masking strategy to train FSKD model. The model is tested under various occlusion probability p to query keypoints. The higher p, the more difficult to predict keypoints. The average PCK scores of *five* subproblems in Animal dataset are reported. The occlusion uses **gray box** in testing.

RGB	Saliency	Patch feature	0%	10%	30%	50%	70%	100%
✗	✗	✗	44.40	42.01	37.33	32.46	27.58	20.08
✓	✓	✗	45.16	44.45	42.64	40.68	38.98	35.94
✗	✓	✓	44.86	42.64	37.69	32.80	27.97	20.14
✓	✓	✓	44.81	44.12	42.06	40.16	37.85	34.53
✗	✓	✗	44.73	42.13	37.50	32.22	27.55	19.56
✓	✗	✗	44.66	43.87	42.08	40.53	39.37	36.42

Table 9: The occlusion handling ability of SalViT based FSKD models trained by various masking strategies. ✓indicates the foreground patch masking is applied over a kind of information (*i.e.* RGB image, saliency map, or CNN patch features); ✗means no masking. Each row shows a configuration of masking strategy to train FSKD model. The model is tested under various occlusion probability p to query keypoints. The higher p, the more difficult to predict keypoints. The average PCK scores of *five* subproblems in Animal dataset are reported. The occlusion uses **average-pixel box** in testing.

RGB	Saliency	Patch feature	0%	10%	30%	50%	70%	100%
✗	✗	✗	44.16	42.45	38.34	34.06	29.91	22.60
✓	✓	✗	44.90	44.43	42.58	40.94	39.32	36.33
✗	✓	✓	45.38	42.99	38.71	34.39	29.83	22.81
✓	✓	✓	44.94	43.85	42.40	40.41	38.41	35.35
✗	✓	✗	44.55	42.75	38.46	34.14	29.30	22.20
✓	✗	✗	44.54	44.03	42.21	40.80	39.20	36.40